

Decant a Carton into Totes SOP

One episode decants one carton into two totes. The dunnage comes off the top, the two tins go into the left tote, the two boxes go into the right tote, each tote is counted against its own label card and then labeled, the base pad comes out of the empty carton, and the carton is folded flat on the desk where it stood.

The order never changes: air pillow, fill sheet, tins, count and label the left tote, boxes, count and label the right tote, base pad, flatten. No unit is lifted until both top pieces of dunnage are in the dunnage bin, no label card is placed until its tote has been counted, and the carton is not flattened until it is bare.

The left gripper works the dunnage, the carton's left half, the left tote, and card A. The right gripper works the carton's right half, the right tote, and card B. Both grippers press the carton flat together in the last step. Neither gripper crosses the middle.

The carton is never lifted, slid, tipped, or turned while it holds anything, and the totes and the dunnage bin stay where they are put. Nothing is turned in the air, no unit is pulled out from between its neighbors, nothing is stacked or wedged in a tote, and no gripper opens anywhere but over the spot the thing in it is going into.

Setup

Complete both checklists before starting an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole desk: the carton at back center, card A at the back left, card B at the back right, the dunnage bin at front center, the left tote at the front left, and the right tote at the front right.
3. The carton interior is visible from above so every lift can be seen leaving it, and the floor of both totes is visible so the units in each one can be counted from the recording.
4. Both arms are at home with grippers open.
5. The desk is bare apart from the carton, the two totes, the dunnage bin, and the two label cards. There are no marks, trays, or fixtures on it.
6. Nothing obstructs the paths between the carton and the front of the desk. The left arm reaches the carton's left half, card A, the dunnage bin, and the left tote. The right arm reaches the carton's right half, card B, and the right tote.

Materials checklist

1. One **open top carton** sitting square on the desk at back center, base flat on the desk, with clear desk on all four sides. It has four walls and no top flaps. Its base flaps are folded flat under it, neither tucked into each other nor taped, so the carton folds flat along its corner creases under an even inward press.
2. Four units lie in the carton in one layer: two **tins** (SKU A) standing lids up in the left half, and two **mug boxes** (SKU B) lying flat in the right half. There is a clear gap between neighbors so a gripper closes on one without touching another, and nothing rests on top of anything else.
3. One **base pad**, a flat card covering the carton floor, lies under all four units.
4. One **fill sheet** of crumpled paper lies spread over the four units, and one **air pillow**, inflated and sealed, lies loose on top of the fill sheet.
5. Two open top **totes**, empty and dry, walls low enough that a unit is lowered in from above without the gripper touching the rim, each with a flat **label ledge** along the top of its near wall. The left tote sits at the front left and the right tote at the front right, both square to the front edge.
6. Two printed **label cards**, stiff enough to keep their shape when pinched at an edge. Card A carries the SKU A name and the number two and lies printed side up at the back left. Card B carries the SKU B name and the number two and lies printed side up at the back right.
7. One **dunnage bin**, open top and empty, at front center between the two totes, with its rim clear of both totes.

Workspace layout

Every zone is a position on the bare desk. Nothing is marked, so each position is held by where the container was set.

- **Carton:** at back center, centered between the two arms, close enough that both arms reach its top without leaning out. It stays on the desk for the whole episode.
- **Card A spot:** the back left corner, left of the carton and clear of it.
- **Card B spot:** the back right corner, right of the carton and clear of it.
- **Left tote:** the front left of the desk, in line with the carton's left half.
- **Right tote:** the front right of the desk, in line with the carton's right half.
- **Dunnage bin:** front center, between the two totes and in front of the carton, clear of both totes.

SKU A is packed in the carton's left half and goes to the left tote. SKU B is packed in the right half and goes to the right tote. No carried unit crosses the midline and no carried unit passes over a tote it is not going into.

Vocabulary

- **Arm assignment:** the left arm works all three pieces of dunnage, the carton's left half, card A, and the left tote. The right arm works the carton's right half, card B, and the right tote. Both arms press the carton flat in Step 8. Neither arm takes the other's zones.
- **Unit:** one sellable item of a SKU. SKU A is the two tins, SKU B is the two boxes. Units of a SKU are identical, so any tin satisfies a tin and any box satisfies a box.
- **Dunnage:** the packing material that protects the units and is never a unit. There are exactly three pieces: the **air pillow** on top, the **fill sheet** spread over the units under it, and the **base pad** flat on the carton floor beneath them. All three go in the dunnage bin and none of them ever goes in a tote.
- **Label card:** the printed card naming a SKU and the number of units its tote must hold. It is read before the count and placed after it. Card A belongs to the left tote and card B to the right tote.
- **Label ledge:** the flat top of a tote's near wall, where the label card rests printed side up, square to the front edge, clear of the tote's opening.
- **Unit grasps:** a tin around its rim wall, lid up; a box across its two side walls, taking the full thickness. The air pillow across its middle, the fill sheet at its near edge, the base pad at its near edge.
- **Lift:** taking a unit or a piece of dunnage straight down onto its grasp, closing, and raising it straight up until there is daylight under it, before anything moves sideways.
- **Lower in:** carrying a unit level over the tote, lowering it straight down until it is resting on the tote floor clear of what is already in, and releasing once it is resting. A unit is never dropped in from above the rim.
- **One layer:** every unit in a tote resting flat on the tote floor, none stacked on another, none leaning on a wall, and none wedged upright against another. The tins stand lids up and the boxes lie flat.
- **Settled:** the thing stays put for 2 seconds after the gripper lifts clear, with nothing rocking, sliding, tipping, or rolling.
- **Count hold:** both grippers clear of a tote and out of the way, the tote and its label card both in frame, held still for 2 seconds so the units in the tote can be counted from the recording.
- **Release point:** over the dunnage bin, over the tote, or over the label ledge the thing in the gripper is going to. A gripper opens nowhere else.
- **Flat:** the carton lying folded on the desk with no wall standing, its base flaps splayed out flat, nothing trapped under it, and no part of it springing back up when the grippers lift clear.

Steps

Steps 1 to 8 are one carton. Every unit is lifted straight up out of the carton and carried in the orientation it will be set down in.

Step 1: Clear the top dunnage

Goal: the air pillow and the fill sheet in the dunnage bin and all four units in view.

- The **left gripper** closes on the **air pillow** across its middle, lifts it straight up, carries it level to the **dunnage bin**, lowers it in until it is resting, and releases.
- The **left gripper** then pinches the near edge of the **fill sheet**, lifts it straight up out of the carton, carries it level to the dunnage bin, lowers it in, and releases.
- Take the sheet in one piece. Never tear it, and never push it down the side of the carton to get it out of the way.

Check: the dunnage bin holds the air pillow and the fill sheet, all four units are uncovered and in view, two tins in the left half and two boxes in the right half, and nothing has been pulled out of the carton with the sheet. If a unit comes up with the sheet, lower it back into the carton in its own half before going on.

Step 2: Decant SKU A into the left tote

Goal: the carton's left half empty and both tins resting in one layer in the left tote.

- The **left gripper** closes on a **tin** around its rim wall, lifts it straight up out of the carton, carries it level to the **left tote**, lowers it in until it is resting on the tote floor, and releases.
- Take the second **tin** the same way once the first is resting and settled, and lower it onto clear tote floor rather than onto or against the tin already in.

Take each tin straight up. Never slide one out from between its neighbors, never move one unit aside to reach another, and never press the tins down into the tote to make room.

Check: both tins rest in the left tote in one layer, lids up, none stacked, leaning, or wedged, and the carton's left half is bare down to the base pad. If a tin has landed on top of another, lift it clear and lower it onto open tote floor.

Step 3: Count SKU A and label the left tote

Goal: the left tote counted against card A and card A resting on its label ledge.

- Read the number printed on **card A** and count the tins in the **left tote**.
- Hold both grippers clear of the tote for the **count hold** with the tote and card A both in frame.
- The **left gripper** pinches the near edge of **card A**, lifts it straight up, carries it level to the left tote, lowers it flat onto the **label ledge** printed side up and square to the front edge, and releases once it is resting.

The card goes down only after the count matches. If the tote holds any number of tins other than the number on card A, leave the card where it is and end the episode as a count mismatch (see Failures that are not violations).

Check: the count in the tote matches the number on card A, the card lies flat and printed side up on the label ledge, readable in frame, and it is clear of the tote's opening and not resting on a tin.

Step 4: Decant SKU B into the right tote

Goal: the carton's right half empty and both boxes resting in one layer in the right tote.

- The **right gripper** closes on a **box** across its two side walls, lifts it straight up out of the carton, carries it level to the **right tote**, lowers it in until it is resting flat on the tote floor, and releases.
- Take the second **box** the same way once the first is resting and settled, and lower it flat onto clear tote floor.

Never tip or tilt the carton to bring a box within reach. If a box sits deep in the carton, come down on it from above.

Check: both boxes rest flat in the right tote in one layer, none stacked, standing on edge, or leaning on a wall, and the carton's right half is bare down to the base pad. If a box has landed on another, lift it clear and lower it onto open tote floor.

Step 5: Count SKU B and label the right tote

Goal: the right tote counted against card B and card B resting on its label ledge.

- Read the number printed on **card B** and count the boxes in the **right tote**.
- Hold both grippers clear of the tote for the **count hold** with the tote and card B both in frame.
- The **right gripper** pinches the near edge of **card B**, lifts it straight up, carries it level to the right tote, lowers it flat onto the **label ledge** printed side up and square to the front edge, and releases once it is resting.

The card goes down only after the count matches, the same way as card A.

Check: the count in the tote matches the number on card B, the card lies flat and printed side up on the label ledge, readable in frame, and it is clear of the tote's opening and not resting on a box.

Step 6: Clear the base pad

Goal: the carton bare and the base pad in the dunnage bin.

- The **left gripper** pinches the near edge of the **base pad**, lifts it straight up out of the carton, carries it level to the **dunnage bin**, lowers it in, and releases.

Check: the carton floor is bare from front to back and wall to wall, the dunnage bin holds all three pieces of dunnage, and nothing has come out of the carton with the pad. If a unit is still in

the carton, put it in its own tote and count that tote again before going on.

Step 7: Flatten the carton

Goal: the empty carton folded flat on the desk where it stood.

- The **left gripper**, closed, comes down on the outside of the carton's **left wall** at its middle, and the **right gripper**, closed, comes down on the outside of the **right wall** at its middle, both level with the middle of the wall.
- Press both walls toward each other in one even motion until the carton folds along its corner creases and lies flat on the desk.
- Hold both grippers on the folded carton for 2 seconds so the creases set, then lift both straight up clear of it.

Both grippers press together so the carton folds in place instead of being shoved across the desk. Never flatten it with one gripper, never press on the rim of a single wall to tip it over, and never lift the carton off the desk to fold it in the air.

Check: the carton lies flat where it stood, no wall standing, base flaps splayed out flat, nothing trapped under it, and it stays flat when the grippers lift clear. If a wall springs back up, press it down again with the gripper on its own side.

Step 8: Confirm the decant and end the episode

- Confirm the left tote holds its two tins with card A on its ledge, the right tote holds its two boxes with card B on its ledge, the dunnage bin holds all three pieces of dunnage, and the carton is flat.
- Return both arms home, clear of the carton, the totes, and the bin, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the two label cards off the ledges and lay them printed side up at their spots, card A at the back left and card B at the back right.
2. Empty both totes and set the four units aside. Set the totes back at the front left and front right, square to the front edge and empty.
3. Take the three pieces of dunnage out of the bin and set the bin back at front center, empty.
4. Open the flattened carton back into a box, square its four walls, and fold the base flaps flat under it without tucking or taping them. Set it square at back center, base flat on the desk, with clear desk on all four sides.
5. Lay the base pad flat on the carton floor, then lay the two tins lids up in the left half and the two boxes flat in the right half, all in one layer with a clear gap between neighbors and nothing resting on anything else.

6. Vary which unit sits where in its own half between episodes rather than rebuilding the same pattern every time. The counts stay at two tins and two boxes and must match the numbers on the two cards.
7. Spread the fill sheet over the units and lay the air pillow loose on top of it. Replace a fill sheet that has torn or an air pillow that has lost its air.
8. Check the carton: it must stand square and fold flat again under an even press on its two side walls. A carton whose creases have gone soft, whose walls will not stand, or that will not fold flat is replaced before the next episode.
9. Replace any unit that is damaged: a crushed box, a dented tin, or a tin whose lid has come off. Replace a torn or unreadable label card, a base pad that will not lie flat, and a tote that is cracked or whose label ledge will not hold a card.
10. Wipe the desk and confirm it is dry and bare. A slick or crumbed desk lets the carton and the totes slide under a press.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** a unit is lifted before both the air pillow and the fill sheet are in the dunnage bin, a label card is placed before its tote is counted, the boxes are decanted before the left tote is counted and labeled, the base pad is lifted before the carton is empty of units, or the carton is flattened before the base pad is out.
- **SOP rule broken:** Steps 1 to 7, the order never changes.
- **Coaching note:** finish the step you are in before you start the next one.

Violation: Arms swapped roles

- **Visible cue:** the right gripper takes dunnage, a tin, card A, or reaches the left tote; or the left gripper takes a box, card B, or reaches the right tote.

- **SOP rule broken:** Steps 1 to 7, each arm keeps its own half of the carton, its own tote, and its own card.
- **Coaching note:** each arm stays on its own side. Nothing crosses the middle.

Violation: Unit in the wrong tote

- **Visible cue:** a tin is lowered into the right tote or a box into the left tote, or a unit is carried across the midline.
- **SOP rule broken:** Steps 2 and 4, SKU A goes to the left tote and SKU B to the right tote.
- **Coaching note:** left half to the left tote, right half to the right tote, every time.

Violation: Dunnage left in the carton

- **Visible cue:** a unit is lifted while the fill sheet still covers part of the layer, the fill sheet is pushed down the side of the carton instead of taken out, or the carton is flattened with the base pad, the sheet, or the pillow still in it.
- **SOP rule broken:** Steps 1, 6, and 7, all three pieces of dunnage come out and go in the dunnage bin.
- **Coaching note:** the carton is bare before it is folded. Three pieces out, every episode.

Violation: Dunnage put in a tote

- **Visible cue:** the air pillow, the fill sheet, or the base pad is lowered into a tote, laid on a label ledge, or left on the desk instead of going in the dunnage bin.
- **SOP rule broken:** Steps 1 and 6, no piece of dunnage ever goes in a tote.
- **Coaching note:** dunnage has one destination. It is the bin.

Violation: Count skipped

- **Visible cue:** a label card is picked up with no still hold on its tote, or the grippers stay over the tote through the hold so the units cannot be counted in frame.
- **SOP rule broken:** Steps 3 and 5, hold still for 2 seconds with the grippers clear and the tote and its card in frame.
- **Coaching note:** stop, get clear, and let the count be seen before you take the card.

Violation: Label card mishandled

- **Visible cue:** a card is laid printed side down, lands in the tote or on top of a unit, hangs off the ledge, is folded or creased, is placed on the wrong tote, or is moved again after it has been laid down.
- **SOP rule broken:** Steps 3 and 5, the card lies flat and printed side up on its own tote's ledge and stays there.
- **Coaching note:** printed side up on the ledge, then leave it alone.

Violation: Units stacked or wedged in a tote

- **Visible cue:** a unit is lowered onto another unit, stands on edge, leans on a tote wall, or is pushed down against the unit already in the tote.
- **SOP rule broken:** Steps 2 and 4, every unit rests flat on the tote floor in one layer, clear of the others.
- **Coaching note:** find open floor before you lower it in.

Violation: Unit dug out from between its neighbors

- **Visible cue:** a unit is slid, pulled, or worked out from between the units beside it, or a gripper moves one unit aside to reach another.
- **SOP rule broken:** Steps 2 and 4, no unit is pulled out from between its neighbors and each unit is taken straight up.
- **Coaching note:** everything is in one layer already. Come down on it and lift straight up.

Violation: Unit taken by the wrong part

- **Visible cue:** a tin is taken by its lid, a box by an edge or a corner, the air pillow by its seal, or the fill sheet or the base pad by a corner rather than its near edge.
- **SOP rule broken:** Steps 1 to 6, each thing has one grasp.
- **Coaching note:** land on the grasp point or reset and take it again.

Violation: Unit dragged instead of lifted

- **Visible cue:** a unit slides along the base pad, the carton wall, or the tote floor still touching it, with no daylight under it in transit.
- **SOP rule broken:** Steps 1 to 6, lift straight up until there is daylight before moving sideways.
- **Coaching note:** the lift is its own motion, not the start of the carry.

Violation: Unit turned in the air

- **Visible cue:** a tin tips off its lid up carry or rolls, a box is flipped or swung during the carry, or a card is turned over between its spot and the ledge.
- **SOP rule broken:** Steps 1 to 6, nothing is turned in the air and each unit is carried in the orientation it will be set down in.
- **Coaching note:** orientation is set at the pick, never in the air.

Violation: Regripped in the air

- **Visible cue:** the gripper shifts, rolls, or seats its hold again on a unit, a card, or a piece of dunnage without setting it down first.
- **SOP rule broken:** Steps 1 to 7, set it down before taking a new hold.
- **Coaching note:** put it down before you take a new hold.

Violation: Dropped instead of lowered

- **Visible cue:** a gripper opens above a tote or the bin and the thing falls the rest of the way in, or a unit is released before it is resting on the tote floor.
- **SOP rule broken:** Steps 1 to 6, release only once it is resting.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the carton, over the desk, over the wrong tote, or over open space, and the thing lands anywhere but at the spot it was going to.
- **SOP rule broken:** Steps 1 to 6, a gripper opens over the spot the thing is going to and nowhere else.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Contents pressed or packed down

- **Visible cue:** a gripper presses, pats, or squashes the units in a tote, the units still in the carton, or a card on its ledge, or holds a unit down after it is already resting.
- **SOP rule broken:** Steps 2 and 4, each unit is lowered until it is resting and nothing is pressed down. The 2 second press in Step 7 is for the carton creases only.
- **Coaching note:** leave it how it lands.

Violation: Placement corrected by shoving

- **Visible cue:** a gripper nudges, pats, or slides a unit or a card into position after releasing it, instead of lifting it clear and lowering it again.
- **SOP rule broken:** Steps 2 to 5, correct a bad placement by lifting the thing clear and lowering it into place again.
- **Coaching note:** lift it and lay it again, or leave it alone.

Violation: Carton moved

- **Visible cue:** the carton slides, turns, tips, or lifts off the desk while it still holds anything, or it is dragged toward the front edge to bring the contents within reach.
- **SOP rule broken:** Steps 1 to 6, the carton is never lifted, slid, tipped, or turned while it holds anything.
- **Coaching note:** reach down into it. Never bring it to you.

Violation: Tote or bin moved

- **Visible cue:** a tote or the dunnage bin is pushed, dragged, lifted, or turned during the episode, or a gripper leans on a rim hard enough to shift it.
- **SOP rule broken:** Steps 1 to 7, the totes and the dunnage bin stay where they are put.
- **Coaching note:** the containers do not move. Only what goes in them does.

Violation: Carton flattened the wrong way

- **Visible cue:** the carton is folded with one gripper, pressed on the rim of a single wall so it tips over, lifted off the desk to be folded in the air, or shoved across the desk while being pressed.
- **SOP rule broken:** Step 7, both grippers press the two side walls toward each other in one even motion with the carton on the desk.
- **Coaching note:** both grippers, both walls, one even press.

Violation: Carton left standing or springing back

- **Visible cue:** the episode goes on with a wall still standing, the carton bowed rather than flat, something trapped under it, or the carton lifted off its spot after folding.
- **SOP rule broken:** Step 7, the carton lies flat where it stood and stays flat when the grippers lift clear.
- **Coaching note:** press the standing wall down again before you lift clear.

Violation: Dropped, collided, or knocked something over

- **Visible cue:** a unit, a card, or a piece of dunnage falls onto the desk or the floor short of its spot, a tote or the bin is struck and shifts, a tin is knocked over in a tote, a card is swept off a ledge, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 7, keep each thing on a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a unit still in the carton or on the desk, a tote unlabeled, a card off its ledge, dunnage outside the bin, the carton not flat, an arm away from home, an arm still over a tote or the carton.
- **SOP rule broken:** Step 8, confirm both totes, the bin, and the flat carton; return both arms home clear of the scene; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Count mismatch:** the carton holds more or fewer units of a SKU than its card lists, or a unit of one SKU is packed in the other SKU's half. Discard the episode and restock the carton to match the cards.

- **Dunnage missing or extra:** a piece is absent when the step begins, or the carton holds a fourth piece the SOP does not name. Discard the episode and repack the carton.
- **Label card missing, blank, or unreadable** when the step begins.
- **Carton will not fold flat** under an even press on its two side walls, its walls will not stand square, or its creases have gone soft. Replace the carton before the next episode.
- **Carton or tote slides** under a correct press or a correct lower in because the desk is slick, wet, or crumbed. Wipe and dry the desk before the next episode.
- **Tote fault:** the tote is cracked, will not sit square, or its label ledge will not hold a card flat. Replace it before the next episode.
- **Defective unit:** a crushed box, a dented tin, a tin whose lid has come off, or a unit that will not rest still on the tote floor. Replace it before the next episode.

Annotation subtasks (from SOP)

1. Move one piece of dunnage from the carton to the dunnage bin
2. Move one unit from the carton into its tote
3. Count a tote against its label card
4. Place a label card on a tote ledge
5. Flatten the empty carton
6. Confirm the decant and end the episode